Hybrid RAG - Report

Conversational AI - Assignment 2 - Group 121

Streamlit URL: [Hybrid RAG Demo · Streamlit]
Github: [msravanthi1625/AI_Projects: AI_Experiments]

Group members:

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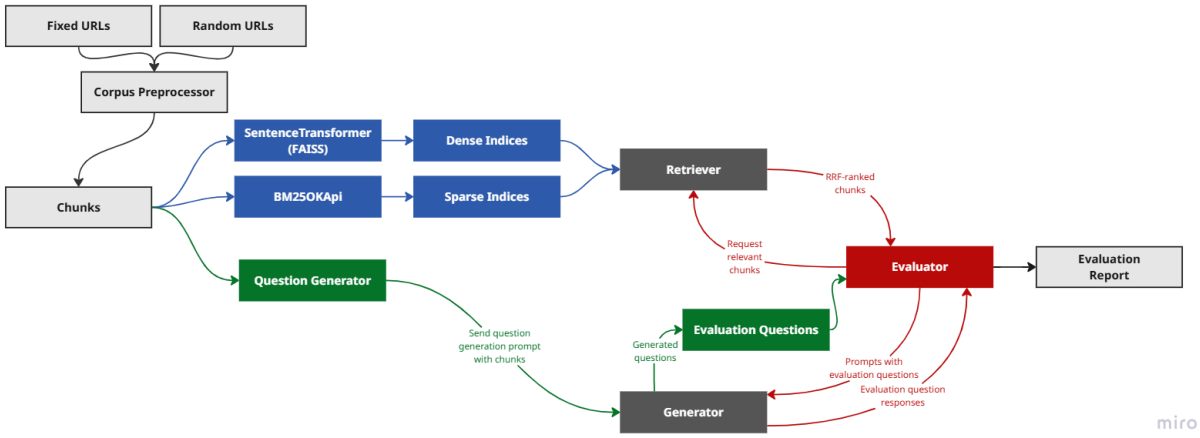
Architecture Diagram



Evaluation Results

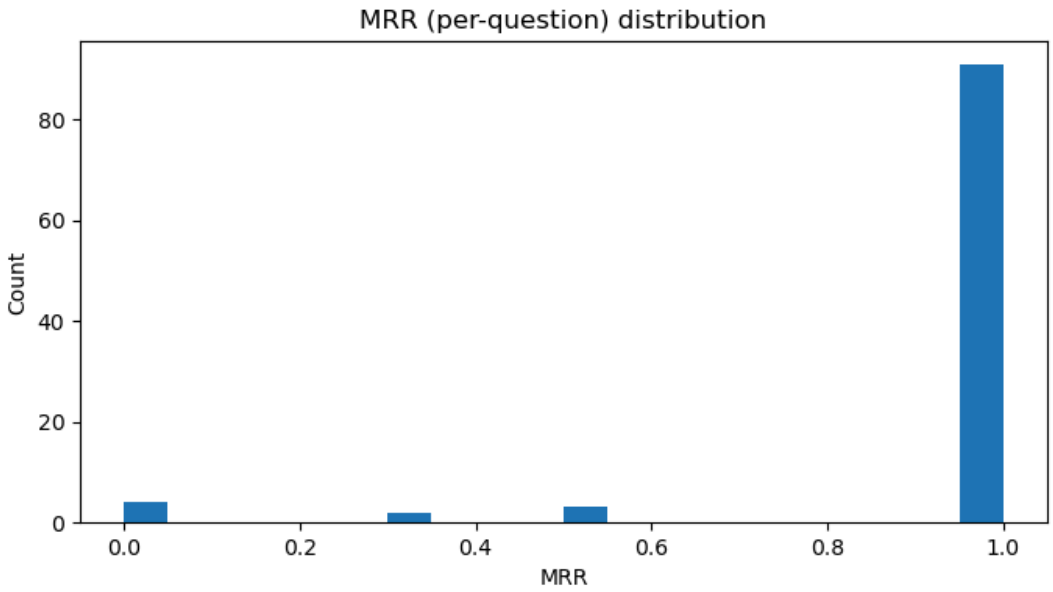
Metric	Metric Category	Value	Metric Selection Reasoning
MRR		0.932	Standard ranking metric that measures the position of the first relevant item, essential for evaluating retrieval system effectiveness

Exact Match	Answer quality	0.32	Directly measures whether the model produces the exact expected answer, critical for assessing QA accuracy and correctness
Recall@10	Retrieval quality	0.96	Evaluates what fraction of relevant documents are recovered in the top 10 results; essential for RAG systems to ensure comprehensive context retrieval
Avg Response Time	Efficiency	0.58s	Measures latency and system responsiveness; critical for user experience and real-world deployment feasibility

This selection of metrics ensures comprehensive and holistic evaluation of our system.

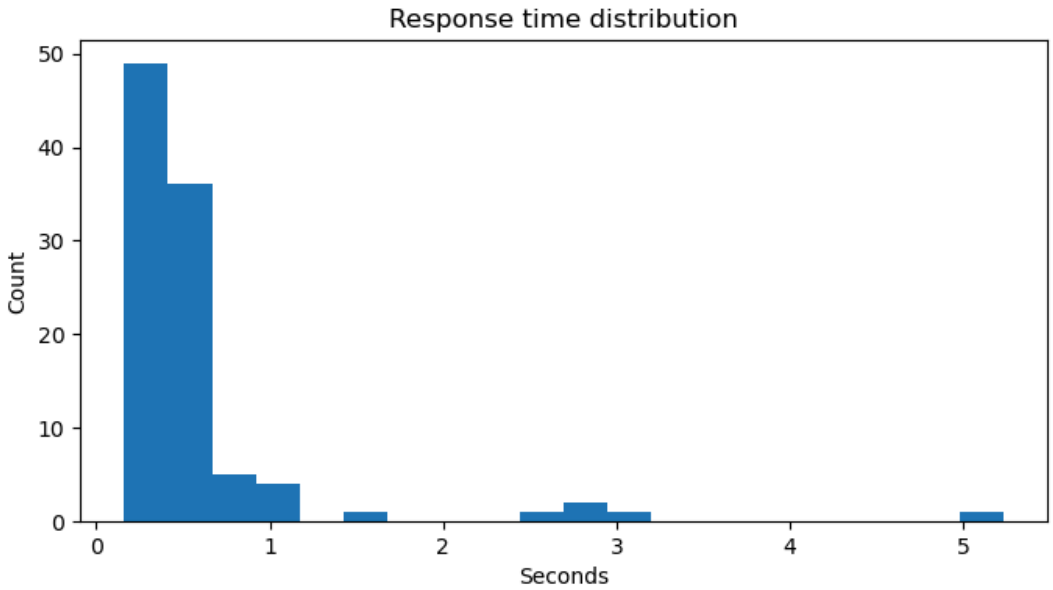
Visualizations

MRR Distribution



Based on this distribution, we can infer that most of the questions' MRR lies closer to the 1.0 mark, pointing to effective context retrieval, thus proving the effectiveness of the hybrid (dense + sparse combined by RRF) indexing approach.

Response Time Distribution



The distribution here shows that most of our queries were evaluated within 1.5 seconds, with a few of our queries taking more than 2 seconds. The maximum amount of time a query took for the response is around 5 secs, which shows how efficient our model is.

Ablation Studies

top_n	dense_k	sparse_k	rrf_k	MRR	Exact Match	Recall@10	Avg Response Time
3	50	50	30	0.9433333333333335	0.33	0.97	1.7215437030792236
3	50	50	60	0.9366666666666668	0.33	0.96	1.8479208087921142
3	50	50	120	0.9366666666666668	0.33	0.96	1.6405984520912171
3	50	100	30	0.9383333333333335	0.32	0.97	1.597994349002838
3	50	100	60	0.9316666666666668	0.33	0.96	1.7887087488174438
3	50	100	120	0.936666	0.33	0.96	1.61170136213

				6666666668			3026
3	50	200	30	0.9383333333333335	0.33	0.97	1.655432550907135
3	50	200	60	0.9316666666666668	0.33	0.96	1.6328124046325683
3	50	200	120	0.9366666666666668	0.33	0.96	1.656341073513031
3	100	50	30	0.94	0.33	0.98	1.5682774233818053
3	100	50	60	0.9366666666666668	0.33	0.96	1.7987668657302855
3	100	50	120	0.9366666666666668	0.33	0.96	1.5922433066368102
3	100	100	30	0.9383333333333335	0.33	0.98	1.6457309484481812
3	100	100	60	0.9316666666666668	0.33	0.96	1.6139427137374878
3	100	100	120	0.9366666666666668	0.34	0.96	1.5817353487014771
3	100	200	30	0.94	0.33	0.98	1.6004781675338746
3	100	200	60	0.9316666666666668	0.33	0.96	1.6163273429870606
3	100	200	120	0.9366666666666668	0.32	0.96	1.7162694644927978
3	200	50	30	0.935	0.34	0.97	1.6657108974456787
3	200	50	60	0.9366666666666668	0.33	0.96	1.6295663022994995
3	200	50	120	0.9366666666666668	0.33	0.96	1.670028603076935

3	200	100	30	0.935	0.33	0.97	1.5469344568252563
3	200	100	60	0.9316666666666668	0.33	0.96	1.6414096999168395
3	200	100	120	0.9366666666666668	0.32	0.96	1.6028388833999634
3	200	200	30	0.935	0.33	0.97	1.646775951385498
3	200	200	60	0.9316666666666668	0.33	0.96	1.5203076958656312
3	200	200	120	0.9366666666666668	0.33	0.96	1.5725844979286194
5	50	50	30	0.9458333333333334	0.28	0.98	1.6258807158470154
5	50	50	60	0.9411666666666667	0.29	0.98	1.7908980298042296
5	50	50	120	0.9411666666666667	0.28	0.98	1.7300587749481202
5	50	100	30	0.9408333333333334	0.28	0.98	1.7583742976188659
5	50	100	60	0.9341666666666667	0.27	0.97	1.7713966941833497
5	50	100	120	0.9391666666666667	0.27	0.97	1.6563507843017578
5	50	200	30	0.9408333333333334	0.27	0.98	1.7555254435539245
5	50	200	60	0.9341666666666667	0.28	0.97	1.8903570151332904
5	50	200	120	0.9391666666666667	0.28	0.97	1.6692635631556128
5	100	50	30	0.94	0.27	0.98	1.7476841497421265

5	100	50	60	0.9416666666666668	0.29	0.98	1.864664831161499
5	100	50	120	0.9416666666666668	0.28	0.98	1.8726015663146973
5	100	100	30	0.9383333333333335	0.27	0.98	1.7504535055160522
5	100	100	60	0.9366666666666668	0.28	0.98	1.8704677438735962
5	100	100	120	0.9416666666666668	0.29	0.98	1.6308416843414306
5	100	200	30	0.94	0.27	0.98	1.6895005321502685
5	100	200	60	0.9366666666666668	0.28	0.98	1.7362376546859741
5	100	200	120	0.9416666666666668	0.29	0.98	1.5920311856269835
5	200	50	30	0.9375	0.27	0.98	1.7176660871505738
5	200	50	60	0.9416666666666668	0.29	0.98	1.8667942905426025
5	200	50	120	0.9416666666666668	0.29	0.98	1.75803578615 1886
5	200	100	30	0.9369999999999998	0.27	0.98	1.7383272957801819
5	200	100	60	0.9366666666666668	0.28	0.98	1.8286876964569092
5	200	100	120	0.9416666666666668	0.29	0.98	1.7128321146965027
5	200	200	30	0.9375	0.27	0.98	1.8694739460945 13
5	200	200	60	0.9366666666666668	0.29	0.98	1.9873379087448 12

5	200	200	120	0.9416666666666668	0.29	0.98	1.7354657936096192

Analysis:
For the query - “What is the name of the group that argued that disco is not a racist movement?“
A Dense Indexing gives - Civil Rights Group - Correct

	rank	rrf_score	dense_rank	sparse_rank	title	url	excerpt
0	1	0.0325	2	1	Disco	https://en.wikipedia.org/wiki/Disco	leading to the decline of disco in the United States
1	2	0.0164	1	None	Disco	https://en.wikipedia.org/wiki/Disco	Disco is a genre of dance music and a subculture t
2	3	0.0159	3	None	Disco	https://en.wikipedia.org/wiki/Disco	the first disco radio show. ==== Early disco culture

A Sparse indexing gives - Disco Demolition Night - Wrong

⋮	rank	rrf_score	dense_rank	sparse_rank	title	url	excerpt
0	1	0.0257	1	48	Disco	https://en.wikipedia.org/wiki/Disco	Disco is a genre of dance music and a subc
1	2	0.0164	None	1	Disco	https://en.wikipedia.org/wiki/Disco	leading to the decline of disco in the Unite
2	3	0.0161	None	2	Hinayana	https://en.wikipedia.org/wiki/Hinayana	a separate sect of the early schools. From

This is observed because the Sparse retriever does exact token matching while the dense Indexing used transformer and performs a semantic match.
Note - The model is trained for “*What is the name of the group that argued that disco sucks was a racist movement?*”

Error Analysis

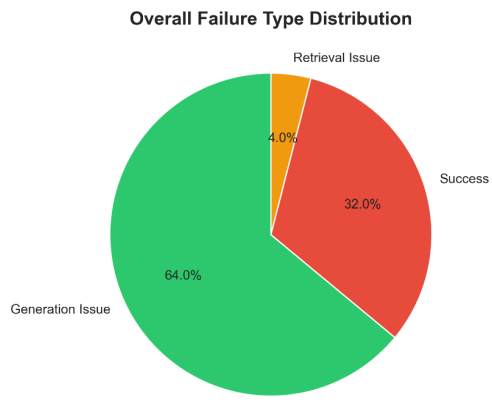
Question Type	Total	Success	Success Rate	Retrieval Issues	Generation Issues
comparative	25	9	36.00%	0	16
factual	25	2	8.00%	3	20
inferential	25	8	32.00%	1	16
multi-hop	25	13	52.00%	0	12

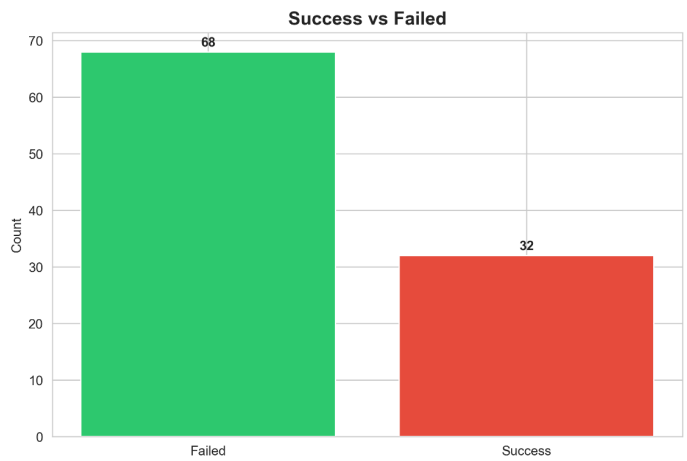
Note: Here, we’ve considered *Success* only when we get the *Exact Match* as the answer, which is a pretty strict condition for success.

As we can see here, most of our issues are *Generation Issues*, pointing to issues with the combination of Generation Prompt and the Generator Model (google/flan-t5-base) itself.

One thing we observed during the course of experimentation is due to the limited context window (512 tokens) of the generator model used, the context chunks in the prompt had to be truncated, leading to occasional truncation of important information from the prompt.

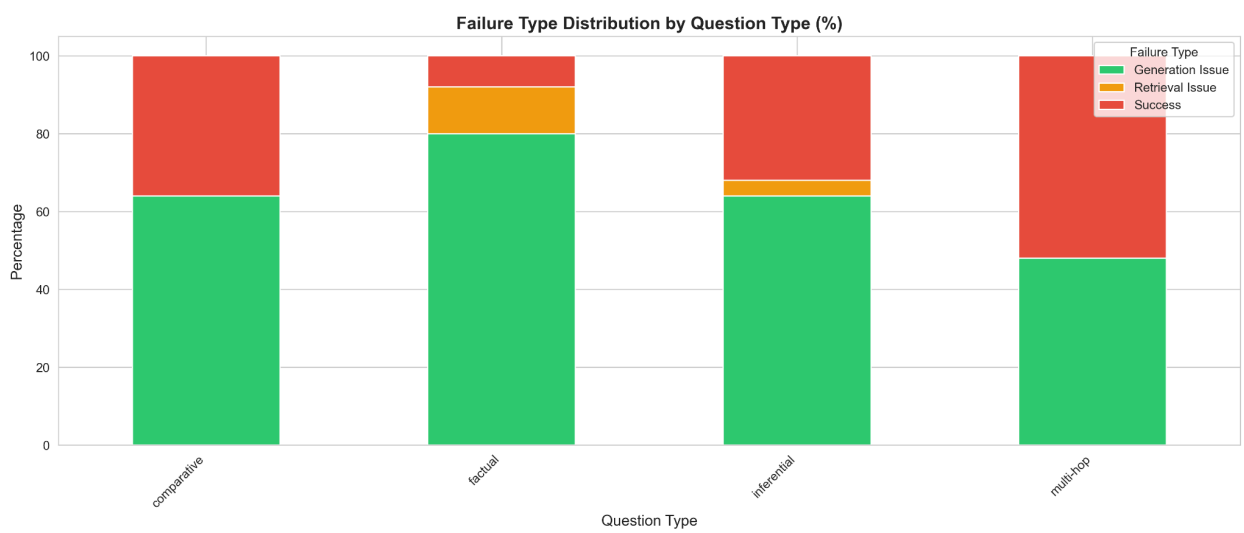
Failure Distribution





Despite the strict condition for *Success* (Exact Match), the success rate was around 32%.

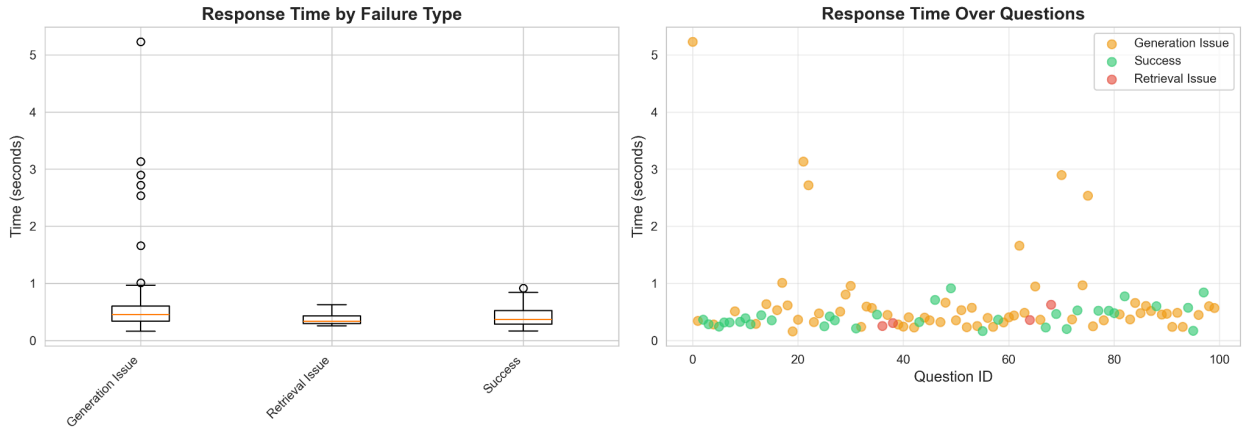
Failure by Question Type



Interestingly, the success rate is higher in *comparative* and *multi-hop* question categories, while being the lowest in the *factual* category.

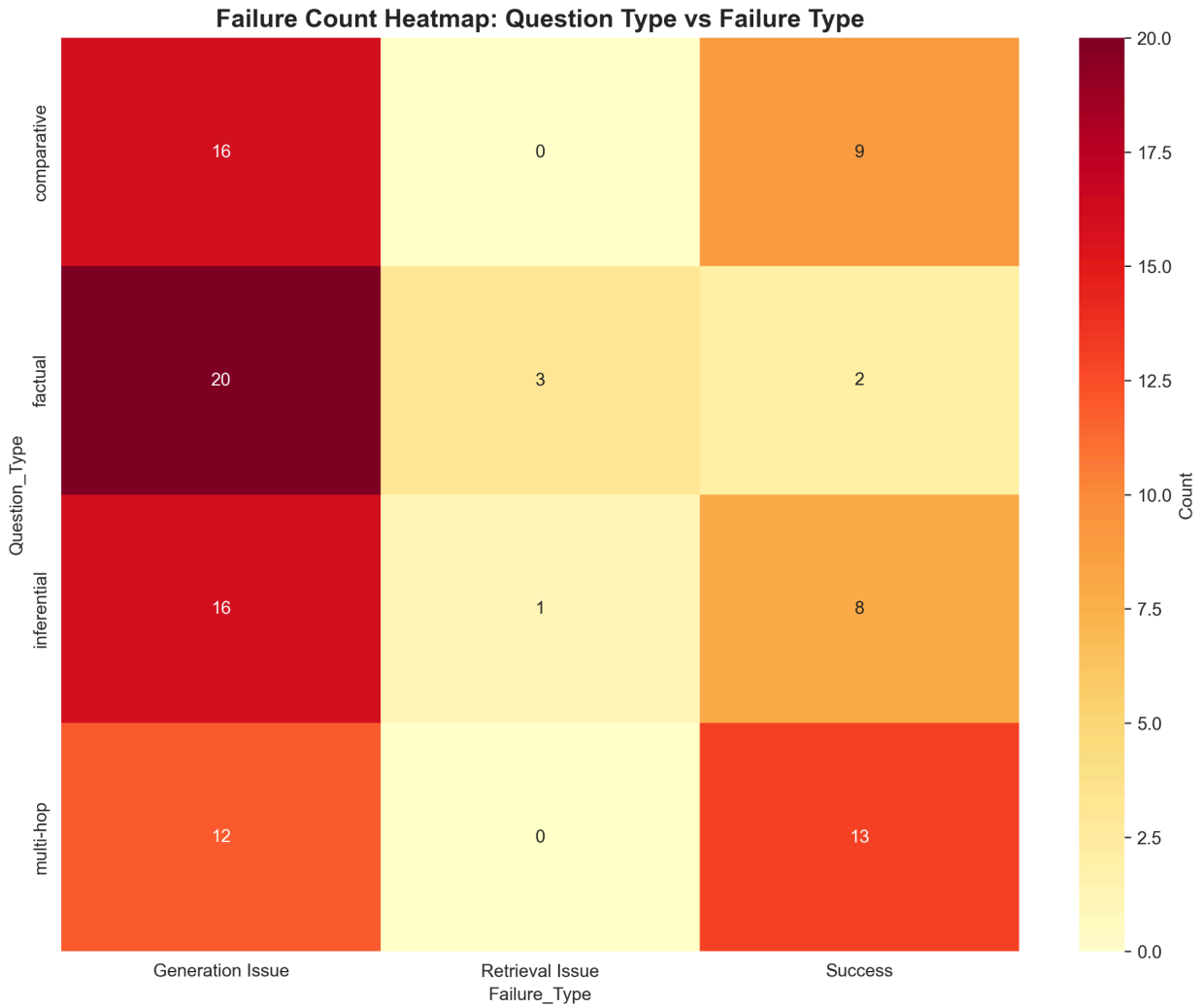
One possible explanation of this uncertainty is the low number of evaluation questions (100). We might see better trends if we use a much higher number of evaluation questions.

Response Time Analysis



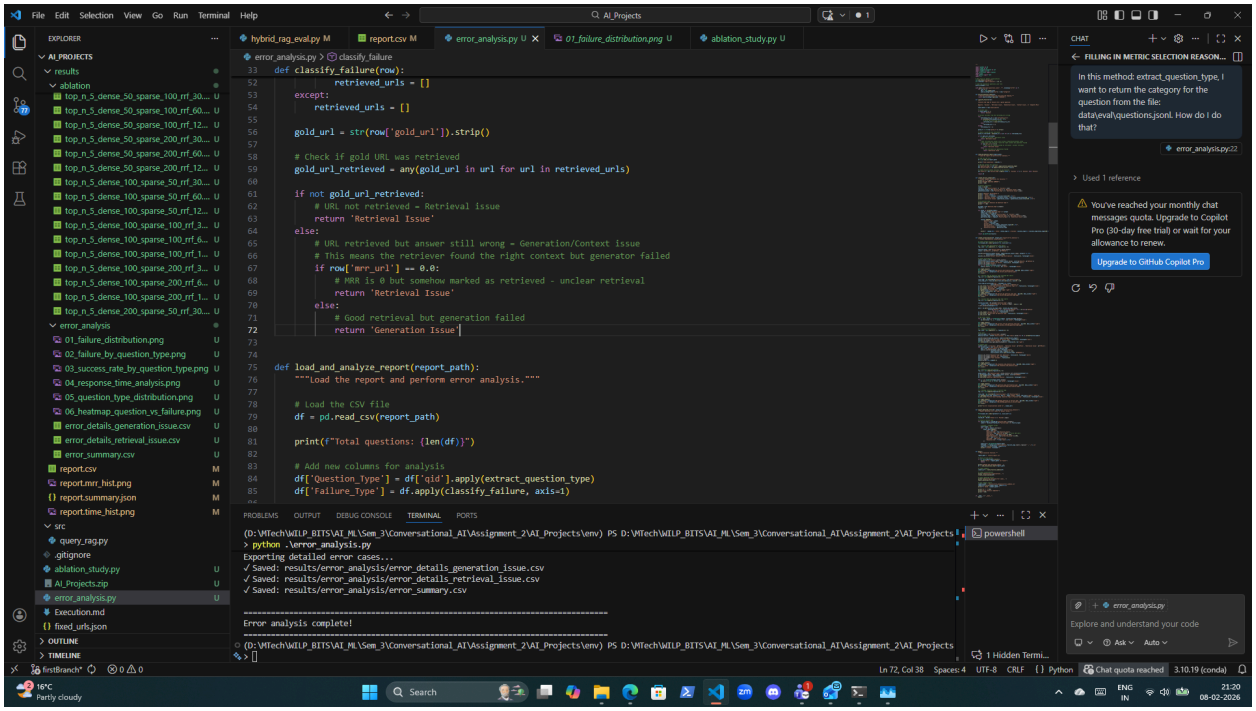
The response times seem to be higher when there's a *Generation Issue* with the response.

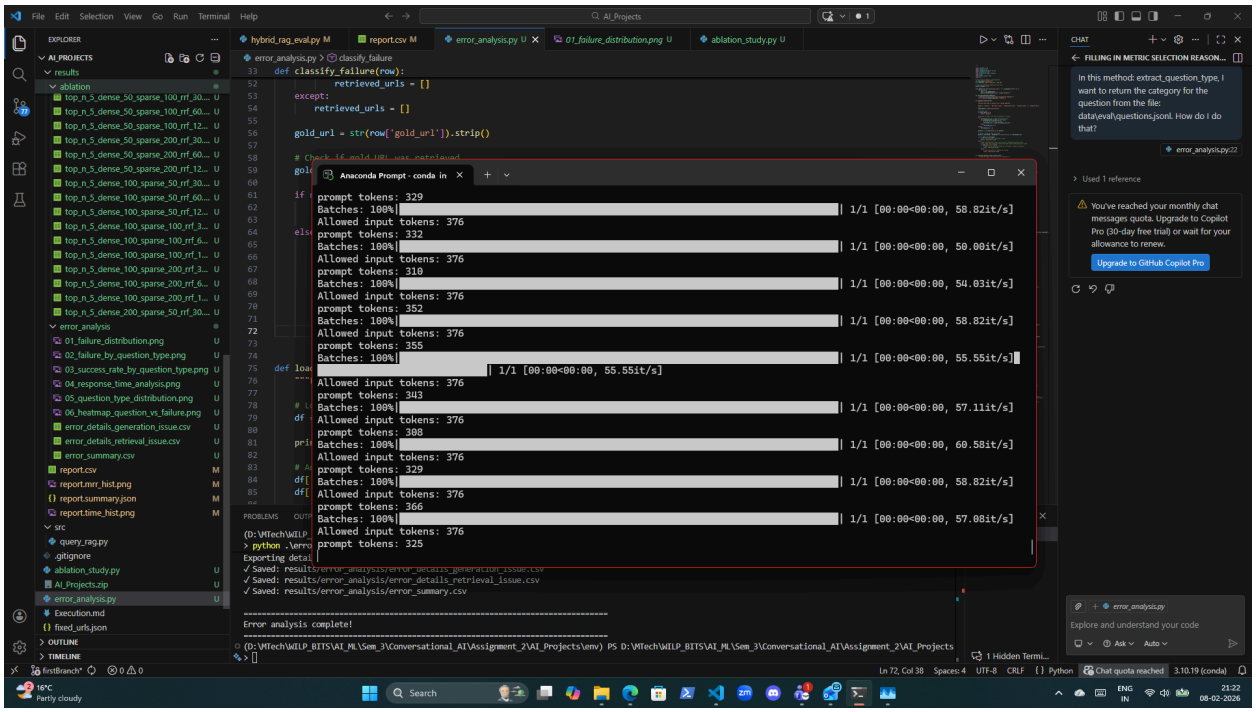
Question Type vs Failure Heatmap

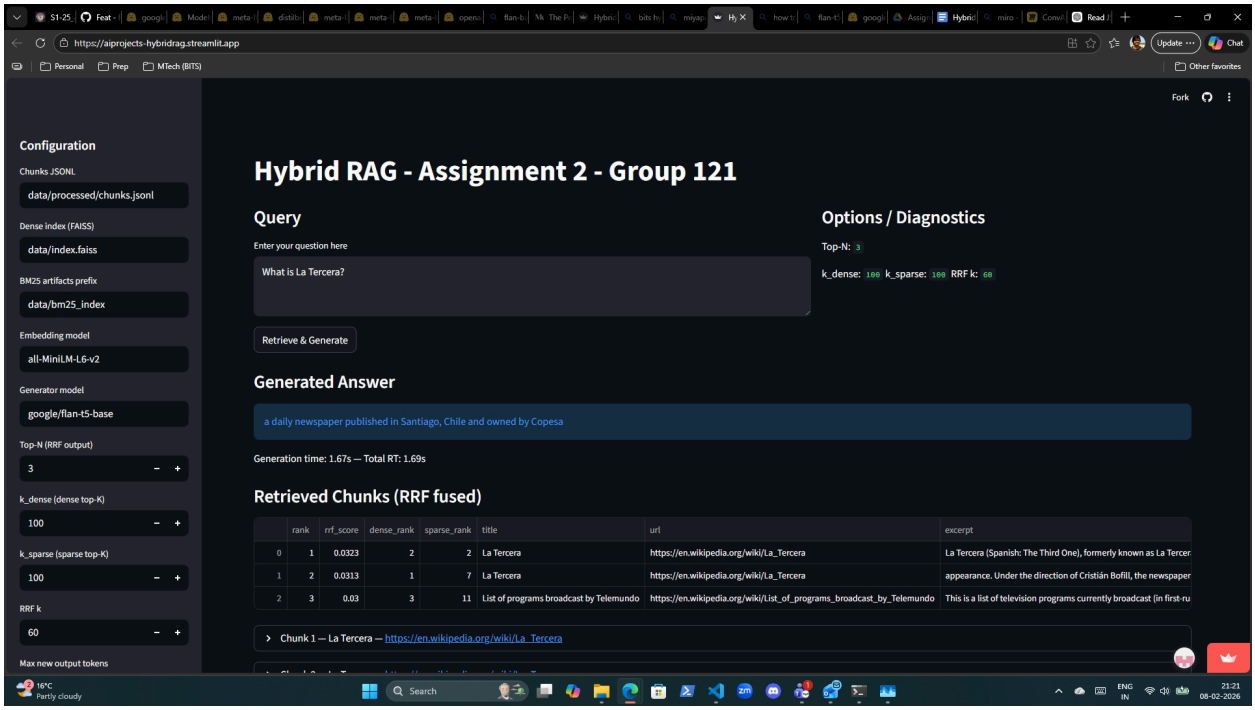


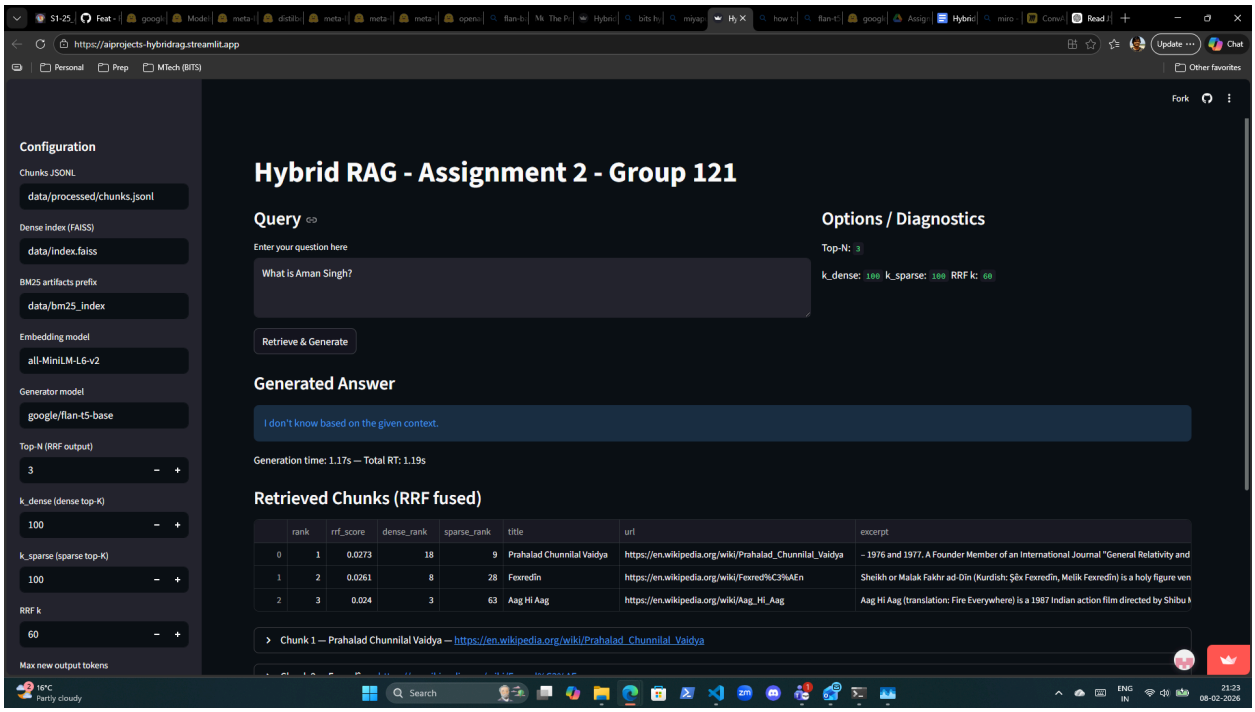
The trickiest category for our model has been the *factual* category.

System Screenshots






https://aiprojects-hybridrag.streamlit.app
Configuration
Chunks JSONL
data/processed/chunks.jsonl
Dense index (FAISS)
data/index.faiss
BM25 artifacts prefix
data/bm25_index
Embedding model
all-MiniLM-L6-v2
Generator model
google/flan-t5-base
Top-N (RRF output)
3
k_dense (dense top-K)
100
k_sparse (sparse top-K)
100
RRF k
60
Max new output tokens
Fork
Hybrid RAG - Assignment 2 - Group 121
Query
Enter your question here
What is La Tercera?
Retrieve & Generate
Options / Diagnostics
Top-N: 3
k_dense: 100 k_sparse: 100 RRF k: 60
Generated Answer
a daily newspaper published in Santiago, Chile and owned by Copesa
Generation time: 1.67s — Total RT: 1.69s
Retrieved Chunks (RRF fused)
rank rrf_score dense_rank sparse_rank title url excerpt
0 1 0.0323 2 2 La Tercera https://en.wikipedia.org/wiki/La_Tercera La Tercera (Spanish: The Third One), formerly known as La Tercer
1 2 0.0313 1 7 La Tercera https://en.wikipedia.org/wiki/La_Tercera appearance. Under the direction of Cristián Bofill, the newspaper
2 3 0.03 3 11 List of programs broadcast by Telemundo https://en.wikipedia.org/wiki/List_of_programs_broadcast_by_Telemundo This is a list of television programs currently broadcast (in first-ru
Chunk 1 — La Tercera — https://en.wikipedia.org/wiki/La_Tercera


https://aiprojects-hybridrag.streamlit.app
Configuration
Chunks JSONL
data/processed/chunks.jsonl
Dense index (FAISS)
data/index.faiss
BM25 artifacts prefix
data/bm25_index
Embedding model
all-MiniLM-L6-v2
Generator model
google/flan-t5-base
Top-N (RRF output)
3
k_dense (dense top-K)
100
k_sparse (sparse top-K)
100
RRF k
60
Max new output tokens
Fork
Hybrid RAG - Assignment 2 - Group 121
Query
Enter your question here
What is Aman Singh?
Retrieve & Generate
Options / Diagnostics
Top-N: 3
k_dense: 100 k_sparse: 100 RRF k: 60
Generated Answer
I don't know based on the given context.
Generation time: 1.17s — Total RT: 1.19s
Retrieved Chunks (RRF fused)
rank rrf_score dense_rank sparse_rank title url excerpt
0 1 0.0273 18 9 Prahalad Chunnilal Vaidya https://en.wikipedia.org/wiki/Prahalad_Chunnilal_Vaidya – 1976 and 1977. A Founder Member of an International Journal "General Relativity and
1 2 0.0261 8 28 Fexredîn https://en.wikipedia.org/wiki/Fexred%C3%AEn Sheikh or Malak Fakhr ad-Dîn (Kurdish: Şêx Fexredîn, Melik Fexredîn) is a holy figure ven
2 3 0.024 3 63 Aag Hi Aag https://en.wikipedia.org/wiki/Aag_Hi_Aag Aag Hi Aag (translation: Fire Everywhere) is a 1987 Indian action film directed by Shibu
Chunk 1 — Prahalad Chunnilal Vaidya — https://en.wikipedia.org/wiki/Prahalad_Chunnilal_Vaidya